



ESTEBAN PINILLA ÁLVAREZ

Mechanical Engineer · MSc Industrial Engineering student (UC3M)

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PROFILE

Mechanical Engineering graduate from the University of Vigo, specialising in Transportation, currently studying the Master's in Industrial Engineering at UC3M. During my internship in the maintenance department of an automotive supplier I automated the calculation of their reliability indicators (MTBF and MTTR). I work with SolidWorks and MATLAB/Simulink, and use Python and artificial intelligence to automate technical tasks. I am looking for mechanical and industrial engineering roles in design, maintenance, production and process improvement.

EDUCATION

Master's Degree in Industrial Engineering (professional qualification) 2026 – present

Universidad Carlos III de Madrid

Bachelor's Degree in Mechanical Engineering · Transportation track Completed 2026

University of Vigo · GPA: 7.0 / 10

- **Bachelor's thesis:** "Design and integration of the steering and transmission system for an amateur go-kart". **Grade: 8.5 / 10.** Simscape Multibody simulation and steering redesign: Ackermann from 253% to 112%, verified in CAD.

EXPERIENCE

Maintenance Department Intern Apr – May 2026

Aludec Inyección · Automotive components (CIE Automotive group)

- Automated the department's weekly **MTBF and MTTR** calculation in Excel with AI assistance (Claude), cutting it from 15–20 minutes to about 2 minutes per run (**87–90% less**) across **1,370** real maintenance records for machines, moulds, peripherals and plant.
- Fixed data quality issues: dates imported as text, inconsistent naming and a week numbering that did not match the company's. Added input validation with 9 fixed categories.
- Wrote an end-user tutorial. The department still runs the tool on its own after my internship ended.

PROJECTS

AI assistant for 2D drawings in SolidWorks 2026 · in progress

- Python engine that generates a part's manufacturing drawing through the SolidWorks API: front view, isometric, optional section and centre marks. Works on all 8 parts of the test set. Built by directing AI agents (Codex, Claude).
- Next phase: automatic dimensioning and an AI layer that interprets requests while the deterministic engine executes, so the language model never touches the geometry.

CanSat Competition · Galicia regional champion 2019 – 2020

- Designed and built, as part of a team, a mini-satellite the size of a soda can.

ENGINEERING SOFTWARE

SolidWorks · advanced
Technical design fundamentals in SolidWorks course, University of Vigo (2025)

AutoCAD · advanced

MATLAB / Simulink · intermediate

CATIA · ANSYS · intermediate

Advanced Excel · Microsoft Office

AI & PROGRAMMING

- **Python** for automation and controlling engineering software through its API (SolidWorks).
- Development with **AI agents** (Claude Code, Codex): specification, planning and verification with automated tests.
- Connecting AI to engineering tools via **MCP** (SolidWorks, MATLAB/Simulink).

LANGUAGES

Spanish · native

English · C1
Cambridge B2 certificate

French · B2
DELF B2

SOFT SKILLS

- Analytical and critical thinking
- Proactive and eager to learn
- Teamwork